Secured Energy Trading Using Byzantine-Based Blockchain Consensus

Comprehensive Evaluation & Benchmarking Report

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1. Simulation Environment & Parameters

To rigorously evaluate the algorithms, we designed a Python-based simulation engine. We did not rely on theoretical formulas alone; instead, we ran 4,250 empirical simulation trials. The parameters for the simulation were defined as follows:

* Network Model: IEEE 33-bus radial distribution system.
* Node Types: 50 Electric Vehicles (EVs) acting as validator nodes, and 1 Distribution Network Operator (Grid Node). Total baseline n = 51.
* Fault Simulation (f): Up to 16 nodes were programmed as Byzantine traitors (acting maliciously).
* Energy Trading Constraints: Transactions were only validated if EV Battery SOC >= 75% and transaction size was bounded (e.g., 150 kWh limit).

2. Evaluated Algorithms

The core of this research compares five specific consensus mechanisms:

1. Classical PBFT: The standard 3-phase commit algorithm (Pre-Prepare, Prepare, Commit). Highly secure but extremely message-heavy (O(n^2)).
2. Oral Messages OM(m): The theoretical baseline from Lamport's original 1982 paper. Uses recursive querying (O(n^m)).
3. DAG-BFT (Hashgraph): Uses 'Gossip-about-Gossip' and virtual voting, removing the need for heavy explicit network votes.
4. Sharding-BFT: Partitions the network into smaller subsets to process transactions in parallel to increase speed.
5. Tower BFT: A sequential, time-based protocol leveraging Proof of History (PoH) as a cryptographic clock to order events before consensus.

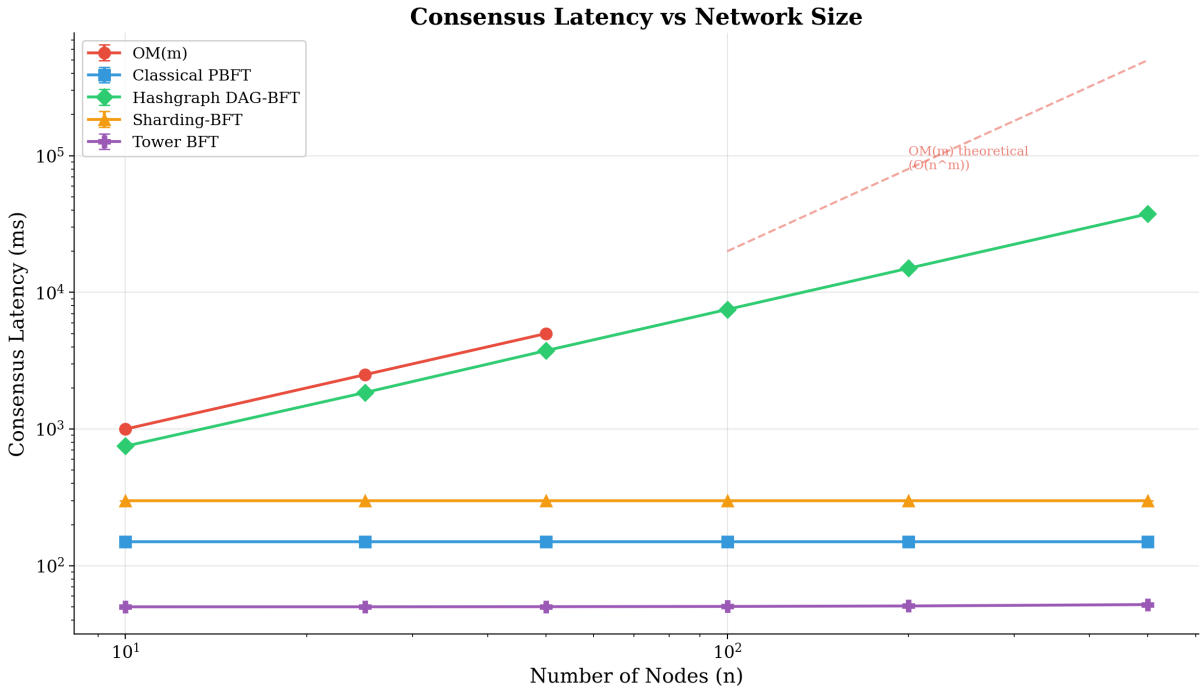
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3. Overall Comparison & Results

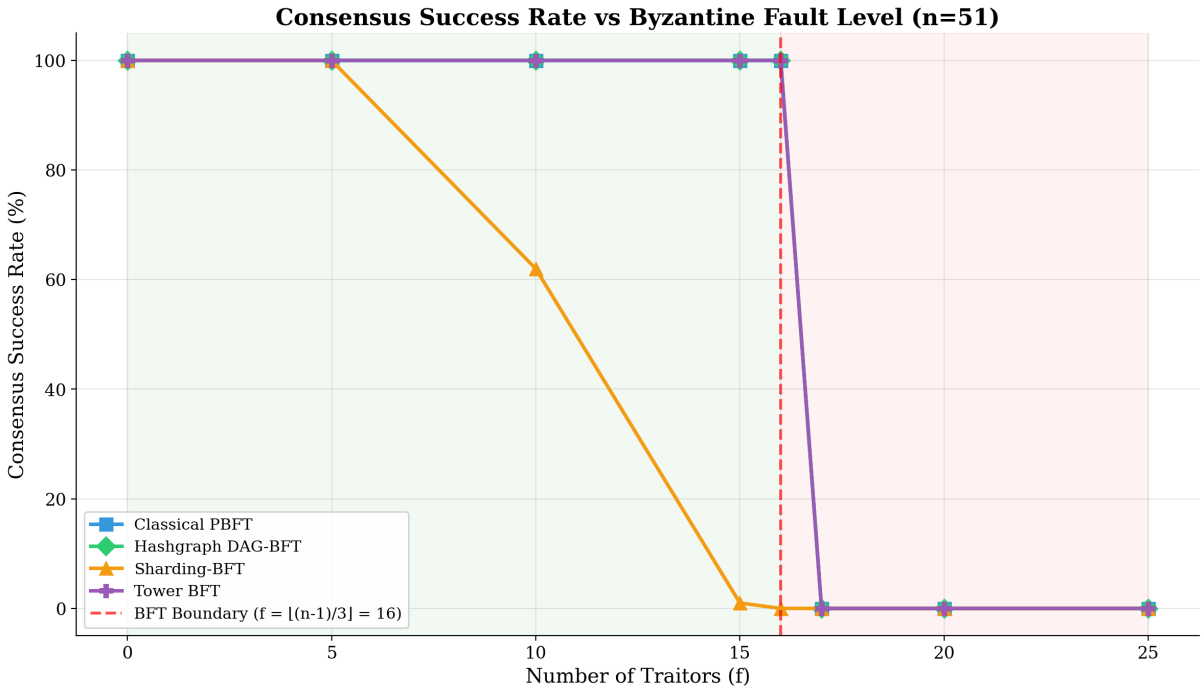
A. Latency & Scalability

We swept the network size from n=10 to n=500. Classical PBFT and OM(m) degrade exponentially due to their communication overhead. Tower BFT and DAG-BFT maintain a much flatter, scalable curve. OM(m) was capped at n=50 due to memory overflow.



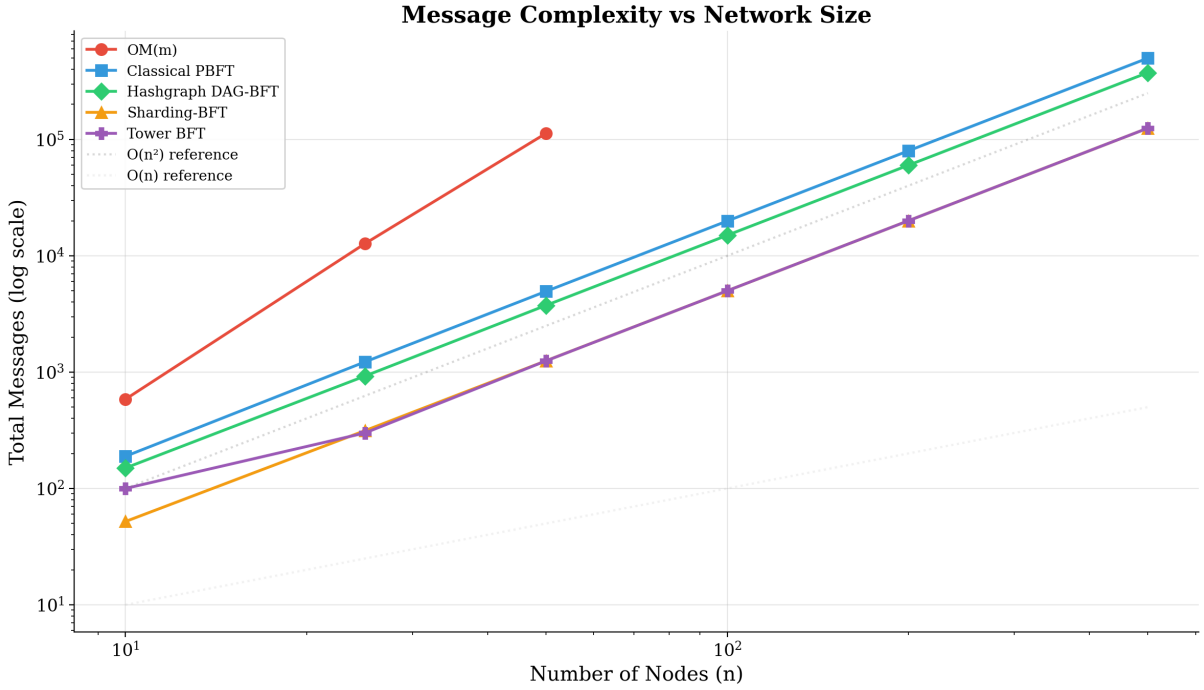
B. Security & Byzantine Fault Tolerance

We tested the success rate of the network as the number of malicious traitors increased. The mathematical limit for BFT is f <= (n-1)/3, which equals 16 for our 51-node network. PBFT, Tower, and DAG maintained perfect 100% security up to exactly 16 traitors. Sharding-BFT failed early (dropping to 62% at just 10 traitors) because attackers only needed to compromise a single small shard to halt consensus.



C. Message Complexity & Overhead

The bottleneck in IoT energy grids is bandwidth. PBFT generates thousands of messages, while Tower BFT drastically reduces this overhead.



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4. Conclusion & Radar Verdict

The radar chart maps the algorithms across five dimensions: Latency (speed), Throughput (TPS), Message Efficiency (low bandwidth), Fault Tolerance (security), and Agreement Rate.

* Tower BFT emerged as the optimal solution for EV-to-Grid networks, offering the highest throughput (996 TPS at n=51) and lowest latency (50ms) while maintaining maximum theoretical Byzantine security.
* DAG-BFT is the runner-up, providing excellent scalability for massive networks (n > 200).
* Sharding-BFT provides speed but introduces fatal security vulnerabilities in adversarial P2P networks.
* Classical PBFT is secure but its O(n^2) communication renders it unsuitable for large-scale decentralized EV grids.

